AI-Powered Digital Marketplace for Upcycling Sri Lanka's Textile Waste- Traceability, recommendations & AI pricing

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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Abstract

A significant amount of the waste generated by the textile industry during manufacturing and production operations is still underutilized because of a lack of transparency and inefficient redistribution systems. One of the biggest challenges in textile waste management is the lack of a reliable system to track material flow and enable informed decisions for reuse or redistribution. Valuable textile waste frequently ends up in landfills, contributing to resource depletion and environmental damage, in the absence of appropriate traceability and intelligent supervision.

This study recommends developing a digital traceability framework in conjunction with an AIpowered recommendation system to improve efficiency and transparency in the redistribution of textile waste. The suggested system will enable the tracking of textile waste batches over the course of their lives by recording crucial attributes such as fabric type, source, quality grade, quantity, and transaction history. This traceability system will provide stakeholders with reliable information about the origins and movement of materials within the platform.

The system will also incorporate advanced recommendation algorithms to assist users in locating suitable reuse options and the most effective ways to redistribute textile waste products. By analyzing material characteristics and historical data, the recommendation model will offer suggestions for potential clients, reuse applications, or recycling techniques that maximize resource use.

The project's expected outcome is a transparent, data-driven platform that enhances decisionmaking in textile waste management and supports circular economy concepts. By offering smart recommendations and improving material visibility, the proposed system aims to reduce the disposal of textile waste, encourage sustainable material reuse, and support ethical production and consumption practices in the textile industry.

Keywords: Textile Waste Management, Traceability System, Artificial Intelligence, Recommendation

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List of abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
ML	Machine Learning
API	Application Programming Interface
CSV	Comma Separated Values
JSON	JavaScript Object Notation
UI	User Interface
REST	Representative State Transfer
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
ESG	Environmental, Social, and Governance
LCA	Life Cycle Assessment
WBS	Work Breakdown Structure
QA	Quality Assurance
DB	Database
EC2	Elastic Compute Cloud
JAAF	Joint Apparel Association Forum
NFR	Non-Functional Requirement
FR	Functional Requirement
UC	Use Case

1. Introduction

1.1 Background and Context

The textile industry is one of the largest manufacturing sectors in the world and plays a significant role in global economic development. However, it also generates a substantial amount of waste throughout its production lifecycle. During garment manufacturing processes such as cutting, trimming, and finishing, large volumes of textile waste are produced in the form of fabric scraps, defective materials, and surplus textiles. A significant portion of this waste is disposed of through landfilling or incineration, which contributes to environmental pollution and inefficient use of valuable resources [1].

In recent years, the concept of the **circular economy** has emerged as a sustainable alternative to the traditional linear production model of "take–make–dispose." The circular economy emphasizes the reuse, recycling, and redistribution of resources to extend the lifecycle of materials and minimize waste generation. Within the textile industry, redistributing manufacturing waste to designers, upcyclers, and recycling organizations can significantly reduce environmental impacts while also creating new economic opportunities [2].

Despite the potential benefits of circular economy practices, the effective implementation of textile waste redistribution remains challenging. One of the primary barriers is the lack of transparency and traceability in the textile waste supply chain. Without reliable information about material origin, quality, and lifecycle data, stakeholders often face difficulties in making informed decisions about reuse or recycling. Therefore, digital solutions that enhance material visibility and traceability are essential for improving textile waste management and supporting sustainable production systems [3]

1.2 Literature Review

Several studies have explored technological solutions for improving sustainability, transparency, and efficiency in supply chains and waste management systems.

One important concept discussed in the literature is **digital traceability in supply chains**. Traceability refers to the ability to track the origin, movement, and transformation of materials throughout their lifecycle. Digital traceability systems have been widely used in industries such as food production, pharmaceuticals, and manufacturing to improve transparency, regulatory compliance, and quality control. In the textile industry, traceability systems can record material attributes such as fabric type, quantity, quality grade, and source information. This allows stakeholders to monitor the lifecycle of textile materials and make more informed sustainability decisions [4].

Another important area of research is the use of **Artificial Intelligence (AI) and recommendation systems**. Recommendation systems are widely used in fields such as

ecommerce, digital marketing, and content platforms to analyze user preferences and provide personalized suggestions. These systems utilize machine learning techniques to analyze historical data, product characteristics, and user interactions in order to recommend relevant items or actions.

In the context of waste management and resource redistribution, AI-based recommendation systems can help identify suitable reuse opportunities for materials that would otherwise be discarded. By analyzing characteristics such as fabric type, quality grade, quantity, and historical demand patterns, recommendation algorithms can suggest potential buyers, recycling opportunities, or alternative reuse applications. Such intelligent systems have the potential to significantly improve resource utilization by connecting waste producers with stakeholders who can effectively reuse those materials [5].

However, despite the increasing adoption of digital technologies and AI in many industries, their application in **textile waste redistribution platforms** remains relatively limited.

1.3 Research Gap

Although existing studies highlight the importance of circular economy principles, traceability systems, and AI-based recommendation technologies, several limitations remain in current textile waste management platforms.

First, many existing textile waste exchange platforms lack **comprehensive traceability mechanisms**. Users often cannot track the origin, quality, or lifecycle history of textile waste materials, which reduces transparency and trust between buyers and sellers.

Second, most current systems primarily function as **basic marketplaces** for buying and selling waste materials. They do not provide intelligent decision-support features that help users identify optimal reuse, recycling, or redistribution options.

Third, the **integration of AI technologies** within textile waste redistribution platforms is still limited. While AI-based recommendation systems are widely used in other industries, they have not been extensively applied to analyze textile waste characteristics and recommend potential reuse opportunities.

Finally, the absence of **transparent material information and supply chain visibility** discourages sustainable reuse practices and reduces confidence among stakeholders.

Therefore, this research aims to address these limitations by developing a digital platform that integrates **material traceability, AI-based pricing prediction, and intelligent recommendation capabilities** to improve transparency and decision-making in textile waste redistribution.

2. Objectives

3.1 Main Objective

To provide a digital platform that combines AI-based pricing, intelligent suggestion capabilities, and textile waste tracking to enhance transparency and facilitate effective decision-making in the redistribution of textile waste.

3.2 Specific Objectives

1. To create and put into place a digital traceability system that keeps track of textile waste batches by recording important characteristics, including fabric type, amount, quality grade, and source information at every stage of the material's lifespan.
2. To create an AI-based pricing model that uses past transaction data and the properties of textile waste materials to forecast their ideal resale value.
3. To put in place an intelligent recommendation system that, based on material characteristics, anticipated pricing, and demand trends, identifies appropriate buyers, reuse prospects, or recycling solutions for textile waste materials.

3. Methodology

Data gathering, system design, model development, and system integration are all part of the structured process that will be used to construct the Traceability and Intelligent Recommendation module. To increase efficiency and transparency in the redistribution of textile waste, the system will include digital traceability, AI-based pricing prediction, and intelligent suggestion algorithms.

4.1 Data Collection and Sourcing

The system's traceability, price prediction, and recommendation features will be supported by both structured material data and past transaction data.

Data sources will include:

- Textile Waste Attributes: Information related to textile waste batches, such as fabric type, quantity, quality grade, color, and condition, provided by sellers during the listing process.
- Historical Transaction Data: Previous records of textile waste sales, including material attributes, selling price, and buyer information.
- Industry Market Data: Publicly available data related to textile waste market trends and resale values.
- A structured dataset will be created in CSV or database format, where each record represents a textile waste batch with the following attributes:
    - Fabric Type
    - Weight / Quantity
    - Quality Grade
    - Source / Manufacturer
    - Material Condition
    - Historical Selling Price

This dataset will be used for training the AI pricing model and supporting the recommendation system.

4.2 Feature Engineering

The process of choosing, modifying, and preparing input variables (features) that the AI models will utilize for price prediction and recommendation tasks is known as feature engineering, and it is an essential part of the machine learning pipeline. By ensuring that pertinent data on textile waste materials is accurately represented, proper feature engineering enhances the performance and accuracy of predictive models.

The input features for the suggested system are made to capture the essential traits of textile waste batches that affect their potential for reuse and market value. These characteristics will either be acquired through other system modules or gathered from the sellers throughout the listing process.

Primary Features

The following primary features will be used as inputs for the AI pricing model and recommendation system:

1. Fabric Type (Categorical) the kind of fabric, including mixed textiles, cotton, polyester, nylon, and viscose. Both the possible reuse applications and the resale price are greatly impacted by this trait.
2. Weight / Quantity (Numerical)
   The overall weight or quantity of the batch of textile waste is often expressed in kilos. Buyer suggestions and price prediction may be impacted by larger numbers.
3. Quality Grade (Categorical / Ordinal)
   The textile waste material is given a quality grade (e.g., High, Medium, Low). Higher resale value and more chances for reuse are typically linked to higher-grade materials.
4. Material Condition (Categorical) shows the physical state of the textile waste, including any unusable excess, leftover cuts, or damaged fabric.
5. Source / Manufacturer (Categorical) determines the source of the textile waste batch, such as the manufacturer or clothing factory that produced the trash.

4.2.1 Target Variable

The primary target variable for the AI pricing model will be:

- **Predicted Resale Price (Numerical)**
  The approximate market value of a batch of textile waste is determined by its attributes and past transaction data.
- Derived variables, such as customer demand patterns and material similarity scores, may also be produced for the recommendation system to provide intelligent recommendations.
- The system will guarantee that the AI models can precisely evaluate the features of textile waste and produce trustworthy pricing forecasts and suggestions for customers through efficient feature engineering.

4.3 Model Development and Selection

This stage concentrates on creating machine learning models to enable the system's intelligent recommendation and AI-based pricing prediction features. To produce precise pricing estimates and pertinent reuse suggestions, the algorithms will examine textile waste characteristics and past transaction data.

AI-Based Pricing Prediction Model

Based on their features, the pricing model seeks to forecast the best resale price for textile waste materials. For this work, supervised regression methods will be employed because the target variable (price) is a continuous numerical value.

The following machine learning algorithms will be explored:

- **Linear Regression**
  A simple statistical model that finds linear correlations between the characteristics of textile waste and the price at which it may be sold. This model will be used as a reference point.
- **Random Forest Regressor**
  Several decision trees are constructed, and their predictions are combined in this ensemble learning technique. Complex interactions between material properties and market pricing can be captured by this approach.

Model Training and Validation

The next stage is to use the prepared dataset to train and assess the machine learning models for price prediction and recommendation. Making sure the algorithms can reliably forecast textile waste pricing and offer recommendations is the aim of this phase.

Data Splitting

The prepared dataset will be divided into two subsets:

4.4

- **Training Dataset (80%)** – Used to train the machine learning models and allow them to learn patterns between textile waste attributes and their selling prices.
- **Testing Dataset (20%)** – Used to evaluate the performance of the trained models on unseen data.

This method helps guarantee that the model can accurately forecast new textile waste listings and generalize effectively.

Model Training

The training dataset will be used to train the chosen machine learning algorithms, including Gradient Boosting Regressor, Random Forest Regressor, and Linear Regression. The models will examine correlations between the goal variable, the resale price of the textile waste, and input characteristics including fabric type, weight, and quality grade.
Similarity-based methods will be used to find connections between textile waste characteristics and previously documented transactions for the recommendation component.

Model Selection

Based on prediction accuracy, dependability, and computational economy, the top-performing model will be chosen once all models have been evaluated. When users submit textile waste materials to the site, the final system will employ the chosen model to produce intelligent suggestions and real-time pricing predictions.

API Development and Integration

FastAPI, a cutting-edge, high-performance Python web framework, will be used to deliver the finished model as a microservice. For traceability, intelligent suggestions, and AI-based pricing, the API will facilitate a seamless connection between the frontend interface and backend machine learning modules.

Key Features:

4.5

- **Endpoint**: A single POST endpoint (e.g., /predict-impact) to handle requests.
- **Request Body**: JSON object containing input features. Example:



Figure 1: Request JSON Format.

- **Response Body**: JSON object containing predictions. Example:

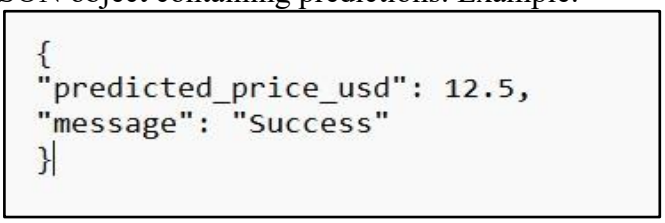


Figure 2: Response JSON Format.

- **Error Handling**

  To assist users in correcting their queries, each endpoint has strong validation for missing or incorrect input data, delivering unambiguous error messages.

- **Documentation**

  Fast API makes integration easier for other team members by automatically creating interactive API documentation (Swagger UI).

4. High-level system architecture

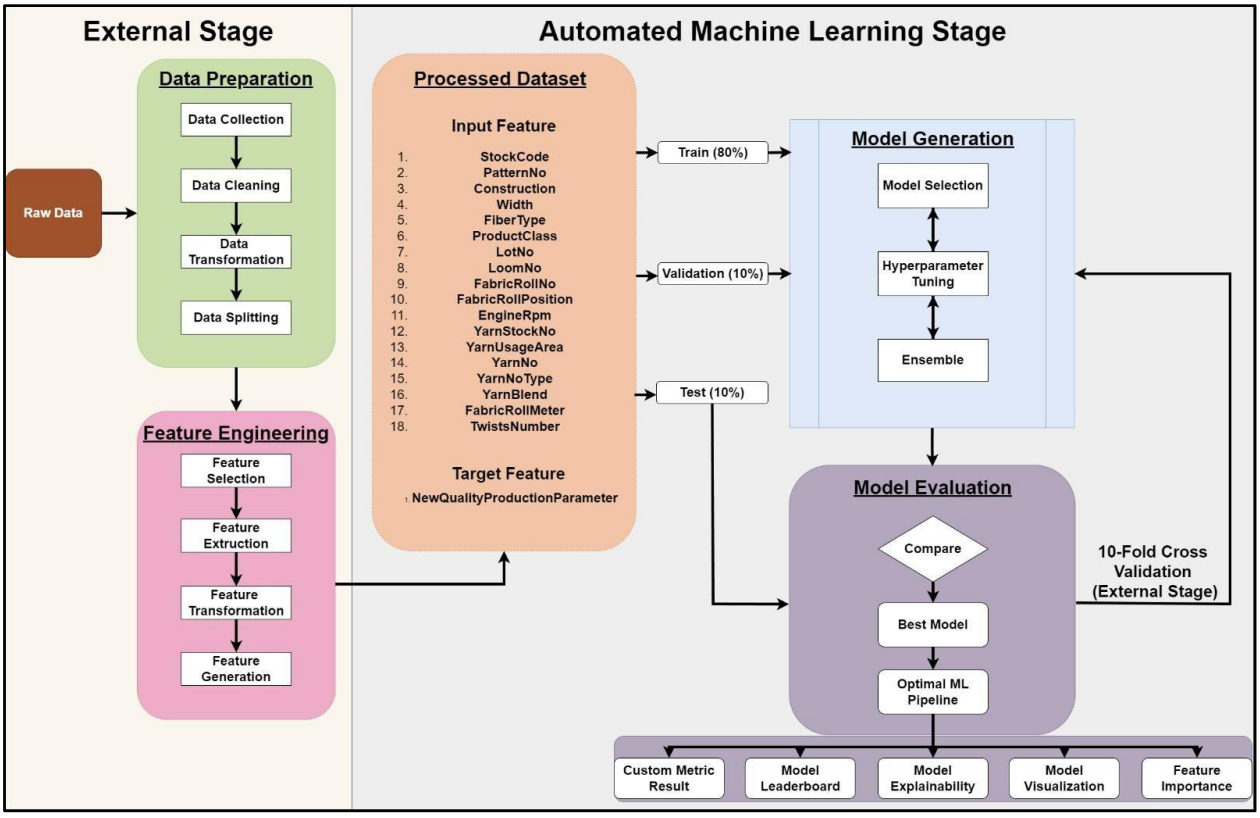


Figure 3: System Overview Diagram

Commercialization Aspects

By utilizing AI-driven pricing, intelligent suggestions, and traceability to generate value for various supply chain partners, the commercialization of the platform aims to transform textile waste management into a feasible business model.

Target Audience

- **Primary:**

- Textile Waste Sellers / Manufacturers AI pricing, suggestions, and traceability data are accessible to factories, tailors, and companies producing textile waste.
- Buyers / Designers / Upcyclers – Businesses or people that source textile waste for recycling, upcycling, or resale.

- **Secondary:**
  - Corporate Sustainability Teams – Can integrate the module as a Sustainability Analytics API for ESG reporting.
  - Waste Management Firms & Platforms – For process optimization, reporting, and traceable textile waste management.
- **Market Space**
  - With a particular focus on the South Asian textile and apparel value chain, the platform addresses the quickly expanding worldwide market for ESG and sustainability reporting software. This specialized market is part of the larger circular economy and sustainability ecosystem, where there is a growing need for clear environmental impact data and workable solutions for managing textile waste sustainably. By meeting these demands, the platform puts itself in a position to assist buyers and producers looking for traceable, environmentally friendly practices and data-driven insights.

Value Proposition:

- **For Manufacturers / Sellers:**
  - Determine the market worth of textile waste using AI pricing, then market it with proven environmental credentials to possibly earn a "green premium."
    Make use of traceability to record the lifespan of materials and enhance sustainability and ESG reporting.
- **For Buyers / Upcyclers / Designers:**
  - Get smart suggestions to find the best solutions for upcycling, recycling, or reuse. Make well-informed sourcing selections and advise customers about the environmental advantages of their goods using data-driven insights.
- **For the Platform:**

- By combining price, traceability, and suggestions into a single service, it sets itself apart from generic markets.
  Establishes the platform as a pioneer in sustainable innovation and circular economy technologies.

How to Promote:

- Content Marketing: Disseminate case studies and white papers that highlight methods, advantages, and practical applications.
- Partnerships: To provide the module to members, work with trade groups like the Joint Apparel Association Forum (JAAF).
- Freemium Model: Provide basic impact estimates at no cost, while premium subscribers may access sophisticated analytics, traceability, and reporting tools.

5. Requirements

5.1 Functional Requirements Table

1: Functional Requirement.

Requirement ID	Requirement Description
FR-001	Role-based access (manufacturer, buyer, admin) will enable users to register, log in, and authenticate.
FR-002	Users will be able to input details on textile waste, such as material type, weight, quality grade, condition, and photos.
FR-003	The system shall validate and preprocess uploaded textile data before sending it to AI modules.
FR-004	Using the AI Pricing Module, the system will forecast the approximate market price of textile waste.
FR-005	Based on textile qualities, the system will offer intelligent suggestions for upcycling, recycling, or reuse.
FR-006	For the submitted material, the system will find possible purchasers or recycling/upcycling partners.
FR-007	All significant data, transactions, and recommendation outcomes must be tracked and recorded by the system in the Traceability Module.
FR-008	In order to offer AI Pricing, Recommendation, and Traceability services, the system will expose Fast API APIs.
FR-009	On the user dashboard, the system will show price, suggestions, and traceability information.
FR-010	Users of the system will be able to export impact and sustainability reports legibly.

5.2Non-Functional Requirements

Table 2: Non-Functional Requirements.

Requirement ID	Requirement Description
NFR-001	Inquiries about price and recommendations will receive real-time responses from the system.
NFR-002 .	The system must be scalable to accommodate several users and data submissions at once.
NFR-003	Sensitive data security, authorization, and secure authentication must be guaranteed by the system.
NFR-003	With appropriate error management and reporting, the system must maintain high availability and dependability.
NFR-005	Clear output visualization and an easy-to-use user interface will be provided by the system.
NFR-006	In order to facilitate future upgrades or the incorporation of additional AI models and metrics, the system must have a modular design.

5.3System Requirements

- **Programming Language:** Python 3.9 or higher
- **Web Framework:** FastAPI
- **Machine Learning Libraries:** scikit-learn, pandas, numpy (for model training and predictions)
- **Deployment:** Docker container; can run on cloud VM (AWS EC2, Google Cloud Run) or integrated into the main backend
- **API Communication:** REST over HTTP/HTTPS
- **Data Storage:**
- Prototype: CSV or SQLite for LCA factor data
- Production: MongoDB or main platform database
- **Version Control:** Git (GitHub or GitLab)

5.4 User Requirements

Although the **Traceability, Recommendation, and AI Pricing module** is a **backend service** and users do not interact with it directly, users of the main marketplace platform rely on its outputs. Therefore, the following user requirements apply:

- Users should have a **basic understanding of sustainability metrics**, such as carbon footprint, water usage, and environmental impact.
- Users should be able to **interpret the Green Score** and other environmental indicators displayed on product listings.
- Users expect the system to provide **accurate pricing, recommendations, and traceability information** through the marketplace interface.
- Users should receive outputs in a **clear, easy-to-understand format**, without needing technical knowledge of the backend processes.

5.5 Use Cases

Table 3: Use Case 1 – Seller Uploads Textile Waste for Pricing and Traceability.

Element	Description
Use Case ID	UC-01
Name	Seller Uploads Textile Waste for Pricing and Traceability
Summary	A marketplace seller uploads textile waste information. The backend module automatically calculates AI-based pricing, generates recommendations, and logs traceability data, which the seller can view on the platform.
Priority	High
Preconditions	• Seller is registered and logged in to the marketplace platform. • Textile waste details (material type, weight, quality grade, condition) are ready for upload.
Postconditions	• Material information is stored in the system and is fully traceable. • Seller can view AI pricing, recommendations, and lifecycle data for their uploaded textile waste. • Data is accessible to buyers or upcyclers.
Primary Actor(s)	Seller / Manufacturer
Secondary Actor(s)	Backend Module (AI Pricing, Recommendation Engine, Traceability System)
Trigger	Seller navigates to the **"Upload Textile Waste"** section on the marketplace platform.

Main Scenario	1. Seller accesses the upload interface on the platform. 2. Seller enters material details: type, weight, quality, condition, and optionally images. 3. Platform sends the data to the backend module via API. 4. Backend module processes the data: • Validates inputs • Predicts market price (AI Pricing)
	• Generates reuse/recycling recommendations (Recommendation Engine) • Logs traceability information (material lifecycle) 5. Platform displays results to the seller: pricing, recommendations, and traceability report. 6. Seller reviews outputs and decides to list the material for sale, recycling, or upcycling.
Extensions	• If the uploaded data is incomplete or invalid, the system asks the seller to correct it. • If the backend module fails, the platform shows a **"Processing Error"** and retries automatically. • If the material already exists, the system warns the seller to avoid duplicates. • If images are not uploaded, the system processes text data but skips image-based assessments.
Open Issues	1. Scalability testing to ensure the system can handle high volumes of uploads. 2. Continuous evaluation of AI pricing and recommendation model accuracy. 3. Standardization of traceability data formats for lifecycle logging. 4. Integration with external marketplaces or recycling platforms.

Table 4: Use Case 2 – View Batch-Level Sustainability Metrics.

Element	Description
Use Case ID	UC-002
Name	User Views Pricing, Recommendations, and Traceability
Summary	Marketplace users view AI-generated pricing, intelligent recommendations, and traceability information for textile waste items, without directly interacting with the backend module.
Priority	High
Preconditions	• User is logged in to the marketplace platform. • Textile waste item has been uploaded and processed by the backend module.
Postconditions	• User can make informed decisions based on Green Score, environmental metrics, AI pricing, and recommendations. • Traceability and lifecycle data are available for the material.
Primary Actor(s)	Buyer / Designer / Upcycler (Marketplace User)
Secondary Actor(s)	Backend Module (Traceability, Recommendation, AI Pricing), Platform Dashboard
Trigger	User navigates to a textile waste item or product listing on the platform.

Main Scenario	1. User selects a textile waste item. 2. Platform requests processed data from the backend module via API. 3. Backend module returns: • Predicted market price (AI Pricing) • Recommended reuse, recycling, or upcycling options • Traceability and lifecycle data 4. Platform displays outputs in an easy-to-understand format, including Green Score and environmental metrics. 5. User interprets the information and decides whether to purchase, reuse, or further explore the material.
Extensions	• If data is unavailable, the platform displays a **"Data Unavailable"** message. • If backend processing fails, the system retries automatically or notifies the user.
Open Issues	• Accuracy of AI pricing and recommendations requires continuous monitoring. • Traceability updates must remain consistent with user actions. • Potential integration with external platforms for extended recommendations.

1. Wireframes

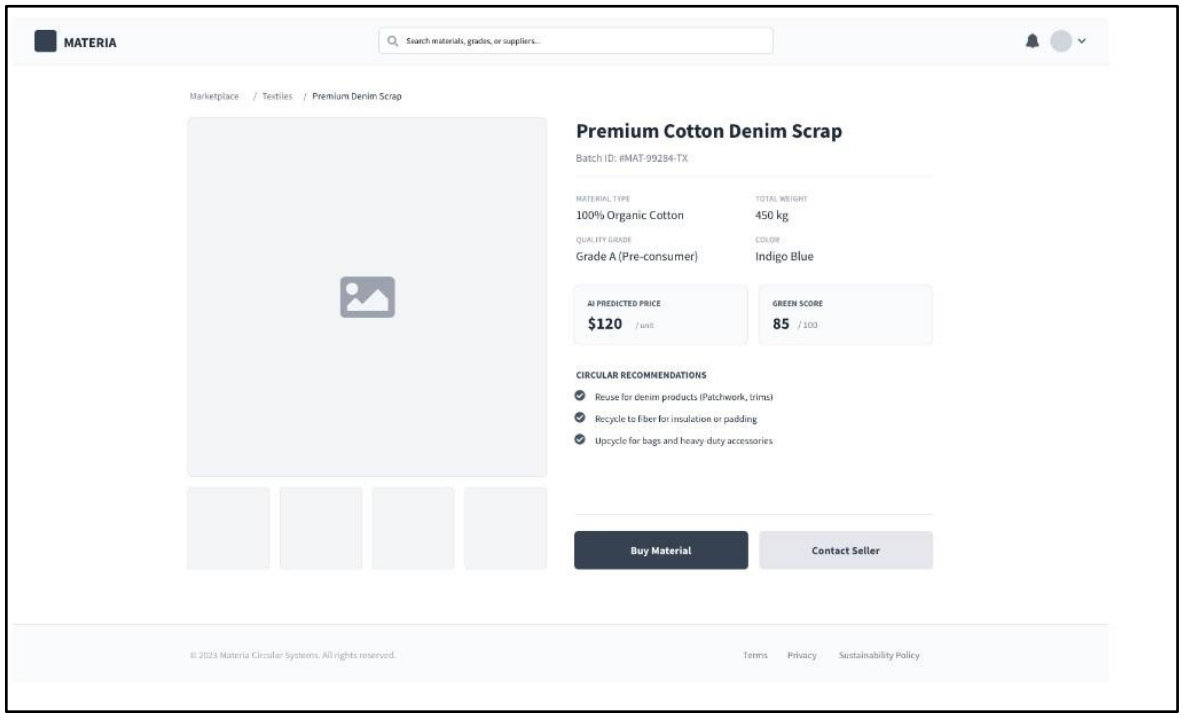


Figure 5: Wireframe01- Ai pricing.

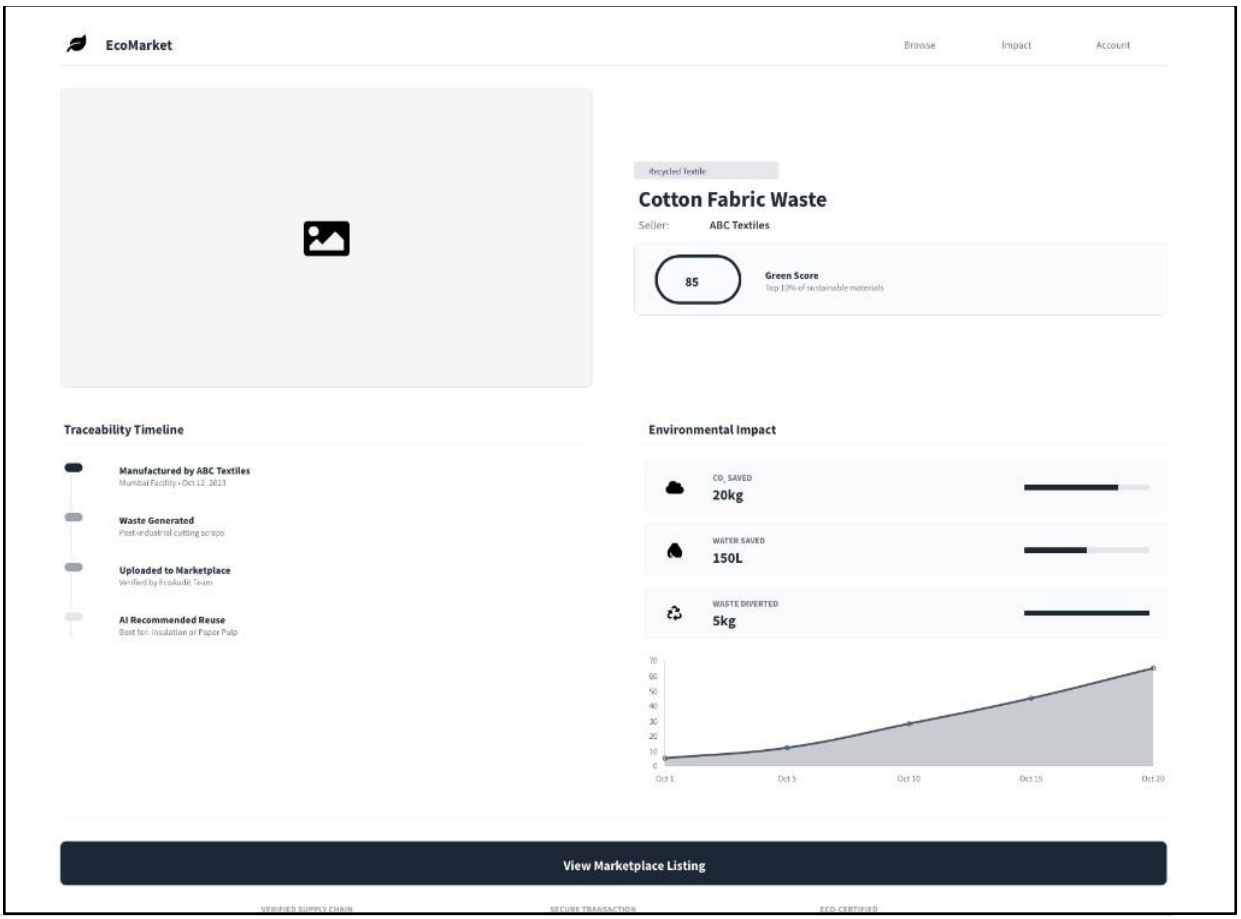


Figure 4: Wireframe02- Traceability.

6. Gantt chart

Table 5: Gantt Chart.

Phase	Task	Start	End	Duration (Weeks)
1. Requirement Analysis	1.1 Gather functional and nonfunctional requirements	Week 1	Week 2	2
	1.2 Identify user requirements and system specifications	Week 2	Week 3	1
2. System Design	2.1 Design system architecture	Week 3	Week 5	2
	2.2 Design database schema & APIs	Week 5	Week 6	1
3. AI Model Development	3.1 Collect and preprocess textile waste data	Week 6	Week 8	2
	3.2 Train AI Pricing Model	Week 8	Week 10	2
	3.3 Develop Recommendation Engine	Week 10	Week 12	2
4. Backend & API Development	4.1 Develop FastAPI microservice	Week 12	Week 14	2
	4.2 Integrate AI modules with API	Week 14	Week 15	2
5. Frontend Integration	5.1 Connect API to the marketplace dashboard	Week 15	Week 17	2
	5.2 Display pricing, recommendations, and traceability data	Week 17	Week 18	1
6. Testing & Validation	6.1 Unit and integration testing	Week 18	Week 20	2

	6.2 Performance, scalability, and security testing	Week 20	Week 22	2
7. Deployment & Documentation	7.1 Deploy module on cloud/production server	Week 22	Week 23	1
	7.2 Prepare user and technical documentation	Week 22	Week 23	1
8. Final Review	8.1 System review, stakeholder feedback, and adjustments	Week 23	Week 24	1

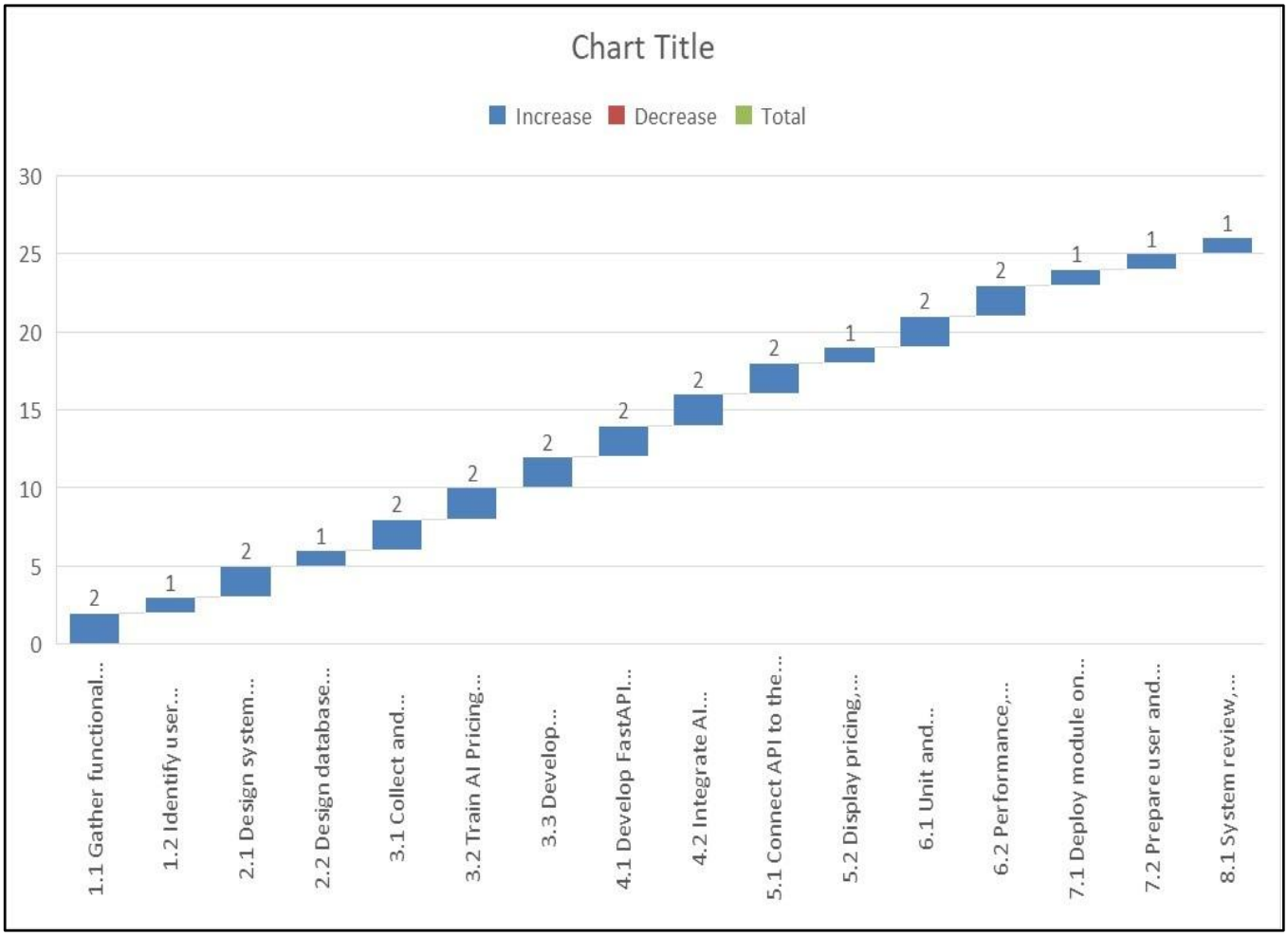


Figure 6: Gantt Chart.

7. Work breakdown structure.

Table 6: Work Breakdown Table.

Level 1 – Project	Level 2 – Major Phase	Level 3 – Tasks / Sub-Tasks
Traceability, Recommendation, and AI Pricing Module Development	1. Requirement Analysis	1.1 Gather functional and nonfunctional requirements. 1.2 Identify user requirements and system specifications
	2. System Design	2.1 Design system architecture 2.2 Design database schema & APIs
	3. AI Model Development	3.1 Collect and preprocess textile waste data. 3.2 Feature engineering pipeline 3.3 Implement baseline model (lookup table) 3.4 Implement and train ML regression models (Random Forest, XGBoost) 3.5 Develop Recommendation Engine 3.6 Develop Green Score calculation logic
	4. Backend & API Development	4.1 Develop a FastAPI microservice with endpoints. 4.2 Implement input validation and error handling 4.3 Write unit and integration tests.

		4.4 Containerize application with Docker
	5. Frontend Integration	5.1 Connect the API to the marketplace dashboard. 5.2 Display pricing, recommendations, and traceability data
	6. Testing & Validation	6.1 Unit and integration testing 6.2 Performance, scalability, and security testing 6.3 End-to-end testing with the main platform
	7. Deployment & Documentation	7.1 Deploy the module on the cloud or production server. 7.2 Prepare user and technical documentation
	8. Final Review	8.1 System review 8.2 Stakeholder feedback and adjustments 8.3 Final project report and presentation/demo

8. Budget and budget justification

Table 7: Budget Table.

Component	Amount (USD)	Amount (LKR)	Justification
Personnel	50	15,000	Estimated value of student development work for AI model, API, and integration.
Software Licenses	11	3,300	Most tools (Python, ML libraries, VS Code) are free; a small allocation for optional tools.
Cloud Services	30	9,000	Cloud hosting or GPU usage for model testing and API deployment.
Hardware	10	3,000	Minor expenses such as external storage or small hardware upgrades.
API & Platform Integration	10	3,000	Resources for integrating APIs with the main system and testing.
Testing & QA	5	1,500	Basic testing tools and debugging resources.
Documentation & Reporting	5	1,500	Printing reports and preparing final documentation/presentation.
Contingency (~10%)	11	3,300	Small buffer for unexpected development costs.
TOTAL	**122 USD**	**39,600 LKR**	The estimated budget for this module of the project.

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Appendix